Harzer Mountain Tuning

Patreon Release Notes

VERSION: V2.5

Long awaited ETK I-Series Support.



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Table of Contents

Housekeeping: 4
HMT Radio: 4
Patreon Tiers: 5
Discord 5
HMT Press Car Config Pack 6
Respect your Elders: ETK I-Series Support 7
HMT Echo Turbocharger Package 8
Turbo Options 8
HMT in House Hybrid Turbo (HAUS) 8
Managing the Hybrid- **HMT DS3 & FMU**: 9
Big Turbos 10
Hollis HBC22-350 10
Hollis H25-660 11
HMT Echo Standalone 11
HMT 6-Speed Swap 13
OEM Gear Sets 13
OEM+ ("Preferred") Gear Set 13
HMT Eden Ratios: 13
HMT SEEK Tri-Cone Carbon Synchronizers 14
Short Shifter 14
Realistic Clutch System 15
Progressive Performance Characteristics 15
Thermal Management 15
HMT Engine Reconditioning Program 16
HMT Ported Head **Error! Bookmark not defined.**
HMT ValveTrain **Error! Bookmark not defined.**
HMT Clutch-fan delete kit 16
Valve Cover 16
HMT Starter 17

Reality: 17
Chassis & Other Supporting Products: 18
HMT Coilovers *18*
HMT Helical Limited Slip Differentials *18*
HMT "Track" Brakes *18*
HMT Radiator *19*
HMT Exhaust System *19*
Aesthetics 20
HeckBlende 20
Yellow Front Fogs 20
Taillights with Rear Fog 20
6000 K Headlights 20
EUDM Lighting Box 20
Paint Matched Airdam 20
HMT Skid Plate 20
Modern ETK Changes 21
Configs: *21*
Modern ETKs: *21*
General Changes: 22
Patreon Content: 23
***New** Patreon Only content:* *23*
Pre-Existing Patreon Content *23*
HMT Eden II NA Intake 23
HMT Carbon Ceramic Brake Pads 23
HMT Twin 2x 8lb Nitrous Bottles 23
HMT Carbon Fiber Half-shafts 23
Pure790 Intake cover 23
Coal Tune for Dusk packages 23
Colorable Filters: 23

Housekeeping:

Welcome Statement:

Hey everyone, and welcome to v2.5 "Respect Your Elders"! With the remaster of the I-Series engine bay, this was inevitable. I've been planning on this since HMT launched nearly 3 years ago but was holding out for better models in the engine bay to work with. Now that it's here, I was finally able to develop content I was happy with both from a visual perspective and of course from a performance aspect. This project has been a fun challenge, creating the "HMT Echo" program with the constraints of older technology. That said, I hope you all enjoy, and thank you from the bottom of my heart for almost 450k downloads here, and over 1M downloads across my entire modding portfolio.

Message To Patrons:

To all current, and future, patrons I want to say thank you. As some of you know, the past few months have been tough for me and having you all for support was a huge help and your patience has meant a lot. I was able to pour what I was feeling into this project, giving me an outlet I desperately needed. So, thank you to all of you, I appreciate you greatly.

-Xpogosam

HMT Radio:

Want to dive deeper into the mindset behind this project? Check out **HMT Radio** on Spotify, a curated playlist that has fueled the development of Harzer Mountain Tuning's content for 3 years. Give it a listen: https://sptfy.com/Qd4a~s



Patreon Tiers:

The Tier benefits are as follows:

- **Supporter ($3.50/month):** Get access to exclusive content upon public repository release.
- **Gold Supporter ($5.00/month):** Get **early access** and exclusive content, up to a month early.
- ***Sapphire Supporter ($7.50/month)***: Includes everything from Gold, plus access to **HMT Annex**, housing products that are either in development, unfinished, or intended to explore new ideas that don't necessarily align with the core ethos of HMT. It is also the best way to support me and my work
    - HMT annex has fallen by the wayside of late, but it still exists. I hope to give it some attention in the near future, but for now the main purpose of Sapphire is for those who want to offer as much support as possible.

Discord

My Discord server has grown into a thriving hub for the community! It's the best place to connect with me, share ideas, and stay up to date on Harzer Mountain Tuning and my other mods. Whether you're into modding, want to provide feedback, or just hang out with like-minded enthusiasts, there's something for everyone.

With over 500 members and counting, the community is filled with great people who love performance tuning and BeamNG. Expect sneak peeks, behind-the-scenes content, and direct engagement with me and other HMT fans. If you would like to join you can follow the link below.

https://discord.gg/aQUVvgnNj9

HMT Press Car Config Pack

Repo Hyperlink or Patreon Hyperlink



With the release of each product line, *Harzer Mountain Tuning* builds a press car to serve as a demo vehicle for media and as a test car for potential clients. These demo cars showcase our latest advancements and are meticulously crafted at our facility in *Halberstadt,* Saxony-Anhalt, to highlight the best of what HMT has to offer.

There are currently **12** cars in total: **seven** 800-Series and **five** K-Series, but this lineup will continue to grow as our offerings expand. Each car showcases one of our packages, our platform-specific chassis setups developed over more than 11,000 in game miles, and custom paint-to-sample colors from various sources, primarily *Pantone* and *Resene*.

The Press Car Config Pack has been Refreshed for **2026**, with new colors, some new parts, and a new Pre-requisite mod, my very own "**TheLightweightCollection**", for some very premium carbon fiber aesthetic and functional bits. The color palette for the 2026 configs has been meticulously curated over the past 6 months, sampling paints from Pantone and "Resene" in New Zealand. This process was carried out in collaboration with a close friend who works professionally as a fashion designer. The result is a palette that is brighter, more expressive, and more varied than previous years, while still remaining cohesive across platforms and performance tiers.

Most configurations received visible refreshment as part of this process. The sole exceptions are **Sagitta** and **Pyxis**, which remain visually unchanged as they're less than a year old. **Cassiopeia** also remains effectively unchanged, though its finish has technically transitioned from RAL Capri Blue to Pantone Classic Blue, a near identical shift made for consistency within the updated palette. Each color was selected not just to stand alone, but to reinforce the personality and intent of the vehicle it belongs to. Additionally, Cepheus and Monoceros have been remarketed as "Concept Studies," not press cars. This moves Cassiopeia and Andromeda to the "flagship" position among the 800-Series press cars.

Respect your Elders: ETK I-Series Support



Founded in 2013, Harzer Mountain Tuning has spent over a decade refining its method for supporting ETK's modern vehicles with the best aftermarket offerings available. With a development process paralleling OEM standards, HMT combines power with precision, engineering upgrades that push the platforms without sacrificing everyday usability. From our early work on the Pure790 and Pure860 turbo packages, to our cult-favorite "Eden II" and Pure560b, our focus has always been clear: elevating the best of ETK with zero compromises.

While our production packages (Pure, Dusk, Eden, etc.) have historically centered around modern ETK platforms, we've long held appreciation for the brand's heritage. Over the years, we've accepted commission projects from owners of the iconic I-Series, a platform that, while aging, is the source of the DNA in ETKs performance engineering, offering a driving experience that ETKs modern lineup draws its roots from.

What began as one-off client projects from our facilities in Halberstadt and Danbury, has since evolved into something greater. The **HMT "Echo**" program is the culmination of years of background R&D, passion projects, and private development work. It has now been brought into the light as a formal product line.

The choice of "Echo" for the name of this product line was a fitting metaphor for the methodology used in development. The past spoke, its message resonating outward, changing over time, and eventually reaching the present. The present received and amplified, sending a version of that same message back to its roots.

HMTs "Echo" Program and our Engine Reconditioning Program brings modern tuning principles to the timeless I-Series, preserving its analog charm while updating critical components with modern tuning technology. Every part of the Echo product line is engineered with the same care, testing, and refinement our users have come to expect, now applied to the platform that paved the way for the cars that HMT made our name refining.

This is not nostalgia, this is **heritage.**

HMT Echo Turbocharger Package

The Echo package in all its variations is the central component of all of HMTs I-Series parts. Basically everything else we offer for the chassis and drive train was built to improve, enhance, and support the Echo turbo kit. As covered prior, this is where we started with customer commissions, and through testing and development with long term customer projects, we ironed out an expansive but curated selection of turbo setups. Whether you're looking for an optimized period correct experience, quick spool with accessible low-end torque, or a big turbo(s) with all the top end you could ask for, we have options available to suit your goals that live up to the exceed all prior standards we have set.



The stock turbo that ETK equipped from the factory suffers from many things, most of which can be attributed to age. The 80s were the dawn of turbocharging in production vehicles, but the turbos and engines of the period lacked the technology to be properly utilized together. The 12-valve head and SOHC of the I-Series engines don't offer enough flow to spin a dated turbo with inefficient internals. Thankfully, in the almost 40 years since the I-Series first hit the market, turbo technology has come a **long** way. HMT has spent several years in development, working both internally and with global supplier "Hollis" to offer **3 turbo options** within our "Echo Turbo System" to address the factory powertrains shortcomings and navigate the engines restrictions in different ways.

Turbo Options

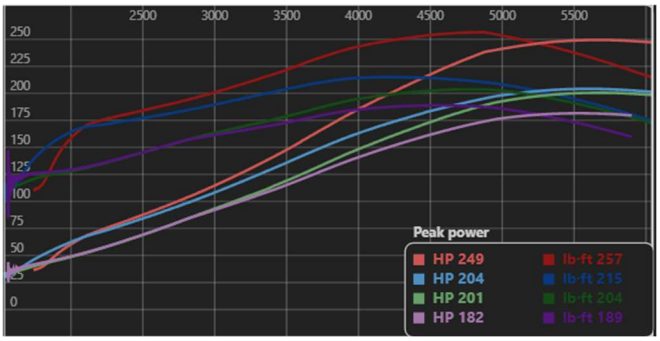
HMT in House Hybrid Turbo (HAUS)

The first step in the Echo ecosystem is HMTs in house developed hybrid turbo. We disassemble, recondition, and upgrade all components of the provided factory core. With improved modern journal bearings, a lighter internal CHRA, and more efficient housings, the HAUS turbo makes much better use of the limited exhaust flow it is subjected to by the SOHC 12-valve head. Rather than choking out at barely more than stock boost levels, like the OEM turbo does, HMTs hybrid will happily spool up to 19psi by redline, a ~50% peak boost pressure improvement. Our hybrid turbo mounts in the stock location on a reconditioned OEM bottom mount manifold, preserving the factory format. It is supported further by HMTs intercooler, intake, and metal charge pipes, all with various aesthetic options of their own to choose from.

The Hybrid option is the perfect choice for an OEM+ restoration looking to bring their I-series up to date while still feeling era appropriate. The improvements made were made specifically to retain the linear boost delivery of the original turbo, while increasing its capability at the same time. The result is a very drivable power plant that pays homage to the spirit ETK built the car with.

Data:

The dyno plot to the right shows several relevant pulls with a color key below*.



- Stock @ 10psi 182bhp/ 189lbft
- **Stock @ 12.4psi (Max) 201bhp / 204lbft**
- **HAUS Hybrid @ 12.4psi 204bhp / 215lbft**
- HAUS Hybrid @ 18.5psi 259bhp / 257lbft

The takeaway is most apparent when comparing the stock turbo at its max boost (12.4 psi) to the HAUS Hybrid at the same 12.4 psi. Peak output is similar, but the HAUS Hybrid carries a higher area under the curve, meaning more torque and power through the midrange. In practice, the turbo spools sooner and responds more quickly to load changes, resulting in higher boost thresholds, better transient response, and thus more usable torque at low rpms.

Where the stock turbo only reaches its maximum output later in the rev range, the HAUS Hybrid spools earlier and holds stronger midrange, showing roughly a ~30 lb-ft gain around 3000 rpm. When boost is turned up well past the stock turbo's max capabilities, the HAUS Hybrid extends the curve further. At 18.5 psi it gains +77 whp and +68 lb-ft at peak while keeping a linear power-delivery fairly similar to the factory turbo, just with more punch everywhere it counts.

**These numbers are measured at the wheel and were provided to us through testing with SGRs Dyno Utility and DSCs Hub Dyno on an otherwise stock 2400tix.*

Drivetrain loss through the AWD system is calculated to be ~16%.

Managing the Hybrid- **HMT DS3 & FMU**:

This is a spiritual predecessor to our modern "DS9" piggyback and is referred to as the "DS3 & FMU." The HAUS hybrid turbo is our entry level and is a more cost-effective option, making the use of our standalone ECU out of place. HMT opted to continue the period correct route with a boost controller and a fuel management unit (FMU) interfacing with the stock ECU to manage the additional boost, a common and reliable method used in cars of this era where flashing a 40 year old ECU is just not practical or efficient. Given the technology, an electronic boost controller in this format is not possible. Rather a manipulation device sits between the wastegates vacuum line and the stock ECU and adjusts the Duty Cycle. The higher the Duty Cycle of the boost controller, the higher the boost the turbo produces. This is the format for user boost adjustment on the DS3 in the vehicle tuning menu.

Big Turbos



When it comes to turbocharging, few names carry the weight that **Hollis** does. For decades they've supplied everything from OEM units for automakers to motorsport-proven hardware, building a reputation for predictable behavior, reliability, modularity, and consistent results across an endless range of platforms and formats. For the Echo program, HMT wanted big turbo options that met or exceed both ETKs and our own quality and performance standards, making Hollis was the obvious call. (okay you caught me I benchmarked a ton of Garret Turbos)

All of the Echo kits that utilize a Hollis turbo are a top mount format, with and standard but optional intake pipe and filter, as well as various color combinations for charge piping, intercooler, and manifold, all available in black and silver, with the charge pipes having the additional option of being colorable too.

These Echo kits require the use of **HMTs Echo Standalone ECU**, a modern and highly capable engine management solution.

Hollis HBC22-350

The first Hollis unit we offer is the HBC22-350, one of the smallest turbos that Hollis offers. The next step upward from the HAUS hybrid, it is a common choice for smaller displacement engine builds that require quick spool with low exhaust gas volume. It's engineered for **fast spool**, strong transient response, and a broad, usable power band, making the engine feel awake the moment you lean into it.

Hollis's uses ball bearing CHRAs with modern compressor and turbine aerodynamics, making our job as the tuner easier, with cleaner power delivery all around. The HBC22-350 is very much a refinement focused option as much as it is an output focused one. It's the go recommendation for owners who value response and longevity more than their need for output.

Due to its smaller size, the HBC22-350 is capable of at most 26 psi of boost, but it makes fantastic use of that boost, delivering it quickly and producing as much power as it is reasonably capable of. That said, the power levels you can expect are greater than the Hybrid and expect mid-300s to low-400s out of a 2.4L with greater torque available at a moment's notice.

Is it clear yet that I benchmarked a ton of Garrett turbos?

Hollis H25-660

The H25-660 is the largest turbo we offer from Hollis's catalog *so far* and is what we recommend for those looking to chase serious power without much tradeoff in response. The H25-660 takes everything great about HBC22-350 and turns it up several notches. Utilizing decades of experience and next generation "H-Series" developments, this turbo is capable of spooling and sustaining high boost pressure across a wide RPM range without choking out at the top end and maintaining a less than expected amount of lag given its output.

This Turbo is right at home for builds on the 2.4L targeting 450-500whp, particularly with our ported heads and cams. This setup does not require stronger internals than the stock "heavy duty" internals equipped to the ttSport vehicles. We find the H25-660 on the 2.4L to be a perfect usable street car, especially with Map Switching.

When paired with the 3.0L, the benefits of the extra displacement are very apparent, and the limits of the stock HD block are found quickly, for which we recommend upgrading to the UHD block through our Engine Reconditioning Program, or keeping boost under 30psi.

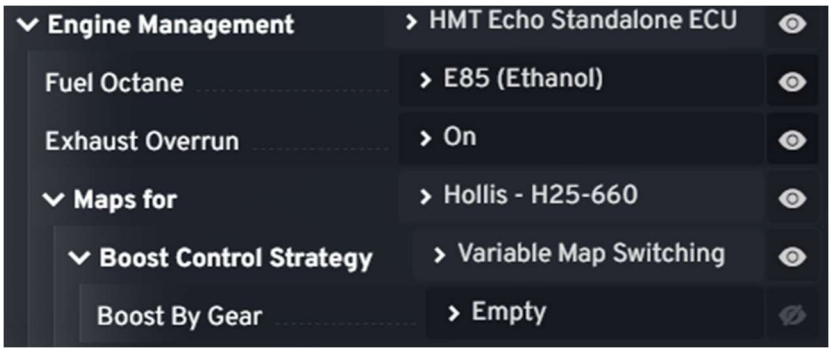
HMT Echo Standalone

The required management solution for the Echo kits, outside the HAUS Hybrid, is our own Echo Standalone ECU. Packed with features and power management capabilities, this is the most complex and most configurable ECU we have ever offered. On the surface it may not seem that different than any other in game race ECU, but with the everything included and the unique formatting and options, that took a lot of effort.

- *Revlimiter cut time*
    - *Instead of a time-based limiter, the UI uses a five-position RPM-drop aggression slider to produce a much sharper, angrier cut, utilizing new internal variables to do so. When paired with our exhaust and the realistic clutch system, the sound on limiter is more aggressive and more unique than any ive heard in game.*
- *Octane Selection*
    - *Presets for 91, 93, and E85 fuel are available, each of which raise the torque limits by a light multiplier. with new E85 logic (ported to Pure560b too) offering the realistic cooling effects of that fuel along with reduced fuel efficiency and increased torque limits.*

Map suites and Boost Control Strategies:

The HBC22-350 and the H25-660 both run different boost levels and as such our standalone is preloaded with options for both,



Within these two preloaded control suites, are two different styles of boost control:

Variable boost control

This should come at no surprise. With balanced wastegate open and limit mathematics, wastegate modulation is under constant control and doesn't act as a binary on/off like many competitors do. This would usually go un-noticed but little things like this add up to a refined and unique package.

Map Switching - Patreon



Yes that's right, the Echo Standalone offers map switching, and unlike the few others who may offer this feature, it's not reliant on "wasteGateOffset" and users can explicitly define each maps target boost, rather than offsetting the boost from the base wastegate spring. No mental math, no guess work. This was surprisingly simple math that I'm shocked hadn't occurred to me sooner, but the refinement is fantastic, and it integrates the same wastegate modulation software as the standard variable boost control. It is also setup for **key binds**, **GUI hooks** to note what map you're in, **a simpleControlButton** in your in-game UI with a number that's constantly visible so you know what map you're in, and an **in-car trigger in the center console** that is connected with the map switching function.

Its not often that I break the fourth wall in these documentations, and when I do its because I want to emphasize the amount of effort, thought, research, trial, and error went into this. The amount of both control and immersion in the map switching feature is something I am incredibly incredibly proud of. It will remain for Patrons for the time being. This whole product list has been in development since BEFORE the I-Series engine refresh, and the map switching and integration alone took several weeks to refine it to the point it's at now, which is time I am only able to justify investing thanks to my incredible patrons.

HMT 6-Speed Swap

As an optional add-on to our Echo I-Series Revival Program, HMT offers a 6-speed transmission swap for the I-Series chassis. This upgrade adapts the modern 6-speed transmission from the 800-Series and K-Series to the I-Series platform. With an included short shifter, users can choose from a variety of options, including multiple OEM gear sets and custom ratios, as well as the first (to our knowledge) realistic clutch system in BeamNG.

OEM Gear Sets

The 6-speed transmission found in modern ETKs includes three OEM gear sets within the same housing: Base, Diesel, and Sport (see Figure 2.1). Offering all three allows users to combine these with the vanilla final drive ratios for a wide range of ratio spacing options.

In addition to the OEM sets, we provide two custom configurations:

OEM+ ("Preferred") Gear Set - **Patreon**

In developing the 6-speed swap, we played around with mixing of ratios from the different available gear sets to see if we could find an OEM+ close ratio upgrade to offer. The result was this, a tailored blend of ratios from all three OEM gear sets, resulting in a close yet streetable setup.

Starting with the already tight Sport gear set, we exchanged the 1st and 2nd gears for the ones from the Diesel gear set to achieve tighter spacing in lower gears. We then were able to adapt 5th gear from the Base gearset and use it as the 6th gear in our tailored gear set, lowering the cruising RPM and adding greater overdrive functionality while still remaining close enough to the Sport 5th gear to not drop out of the power band.

- 1st & 2nd : Diesel set
- 3rd , 4th & 5th : Sport set
- 6th : Base 5th gear used as new 6th

This is an added cost upgrade in the 6-Speed swap package as it requires transmission disassembly rather than just an install but is still a cost-effective method of achieving a close ratio feel.

HMT Eden Ratios - Patreon

Developed for Eden II in 2023, this set features extremely close ratios and "Zerol" bevel gears (a hybrid of straight-cut and helical designs). The result is optimal top end power-band utilization and a distinctive racing-inspired sound that's mild enough to not give you a migraine on the street. Still not the most street-able but that's not the point, it excels in track use

HMT SEEK Tri-Cone Carbon Synchronizers

A first in BeamNG, the modern 6-speed swap comes equipped with upgraded tri-cone carbon synchronizers, developed under HMT's "SEEK" product line. **Developed first and foremost for controller players**, these synchronizers are engineered to handle faster shift speeds enabled by the short shifter and the upgraded clutches while maintaining smooth, reliable engagement across all gears.

Traditional brass synchronizers found in older transmissions struggle with the combination of quick shifts and higher torque loads. The SEEK tri-cone carbon units address this with a three-cone design that distributes clamping force more evenly, reducing wear and improving synchro life even under aggressive use.

This upgrade is particularly beneficial when paired with HMT's realistic clutch system, as it allows for confident, fast shifts without the gear grinding that plagues BeamNG manual transmissions when using controllers. The tri-cone design also provides more forgiving synchronization during rev-matched downshifts and partial-clutch upshifts.

The SEEK synchronizers are standard equipment in the HMT 6-speed swaps and represent significant refinement over both the original I-Series transmission hardware and the standard 800-Series and K-Series synchronizers.

Short Shifter

The short shifter integrated into the base 6-speed swap provides approximately 0.11 seconds of shift time reduction compared to stock I-Series transmissions. This benefit is present regardless of clutch selection and represents the mechanical advantage of the modern linkage and updated SEEK synchronizers.

All clutch options support game controller use thanks to the refined SEEK synchronizer settings, which allow for successful shifts even with partial clutch depression, a **significant improvement over vanilla BeamNG manual transmission behavior.**

Realistic Clutch System - Patreon

Another first in BeamNG, the HMT 6-speed swap includes an optional realistic clutch system. When enabled, clutch slip becomes a factor in putting power to the ground, and the proper clutch needs to be selected to handle your target power level. As a bonus, clutch behavior between gears and under throttle feels more authentic, with on and off load RPM variation occurring for more personality and greater immersion.

This system is optional, and the 6MT swap defaults to a "vanilla" slot entity. Use it if its something you're interested in as it's a fun bit of realism. I kept it for Patreon as it's a feature that I'm not sure everyone will understand or accept, whereas I know my Patrons vibe with stuff like this.

Progressive Performance Characteristics

Beyond torque capacity, each clutch option offers progressively sharper engagement characteristics and faster shift times. The baseline short shifter already provides improved shift speed over the stock I-Series transmission, but performance clutches build on this foundation with quicker engagement and additional shift speed reductions.

- *OEM (430 Nm):*
  - Modern replacement unit sourced from ETK. Smooth engagement with moderate shift speed, perfect for daily driving and enough capacity to handle HMTs hybrid turbo. Represents a significant leap in clutch technology over 80s-90s I-Series clutches.
- *HMT GRAB Stage 1 (640 Nm):*
  - Performance street clutch with noticeably quicker engagement and reduced shift times. Ideal for spirited driving without sacrificing streetability. Faster engagement allows for more aggressive driving while maintaining civility in traffic.
- *HMT GRAB Stage 2 (920 Nm):*
  - Aggressive street/track setup. Significantly faster shifts and firmer engagement. Best suited for high-performance street use and occasional track days. Clutch pedal feel becomes more direct and engagement window tightens considerably, a personal favorite.
- *HMT GRAB Stage 3 (1200 Nm):*
  - Full race-spec clutch with lightning-fast engagement and minimal shift delay. Compromises low-speed smoothness for maximum performance. Advised for track-focused applications only. Engagement is nearly binary, and modulation becomes challenging at low speeds and from a stop.

Thermal Management

Utilizing the wrong, or not strong enough, clutch can produce slip. Excessive slip, especially during launches or while overloaded, generates heat that can reduce clutch efficiency or cause permanent damage over time. Higher-spec clutches feature improved cooling capacity and can withstand much more aggressive use before reaching critical temperatures. In our experience we haven't been able to

HMT Engine Reconditioning Program (ERP)

As part of the Echo product line, and as a means of better supporting the platform, HMT offers a rebuild service in house at our facilities in Halberstadt and Danbury. The engine is removed, stripped, cleaned and rebuilt to OEM specifications. While its apart we are able to further customize the engines components offering a plethora of options:

HMT Ported Head

Improves output but up to ~11-12% linearly over the RPM range, as well as integration of improved exhaustPower to the turbo as a result of freer flowing exhaust gases into the turbine wheel. In practice this means that gains for head porting scale naturally with the rest of the build, in hand with the changes to the bottom end, camshafts, and intake/turbocharger selection, as outlined further below:

Engine Long Block -Split

The system for internals drops the "Engine Long Block" slot of the vanilla engine and is now structured with slots of Displacement>Internals as a subslot of the HMT ERP. Select from the OEM 2.4L or 3.0L displacements and then select your bottom end. This system splits the components of the vanilla long-blocks between two different entities: The bottom end, and the valvetrain.

Bottom End controls torque limits and rotating mass. Additionally, the Stage 1 and Stage 2 bottom ends control the compression ratio, accounting for a portion of the power gains from the vanilla "Stage" long blocks as well, with a 7-11% power bump depending on which block.

Camshafts and Valvetrain (available only within the ported head) account for the rest of the bump in power that the vanilla Stage 1 and Stage 2 blocks offer. This means that vanilla power levels and anywhere in between become much more configurable

Bottom End

This is the internals, controlling factors like maximum RPM, maximum torque, rotating assembly mass, and internal frictions, as well as controlling the compression ratio on the "new" Stage 1 and Stage 2" bottom ends, which account for a portion of the vanilla S1 and S2 long blocks power multiplier, resulting in a bumper in power of 7% and 11% across the whole rev range respectively. The large remainder of the gains associated with the vanilla long blocks are now the responsibility of the new camshafts.

HMT Clutch-fan delete kit

Our most popular add-on, deletion of the clutch fan frees up rotational engine mass and removes a common high RPM failure point. Pair with HMT Radiator to delete radiator shroud

Valve Cover

Another added bit of configurability is the swapable valve covers, with options of Black (OEM) as well as red and navy blue wrinkle coat. Within that is the option of a much nicer looking oil cap taken from the new engine models on the Miramar

HMT Starter

Lastly, the Starter is now a swapable component, with an optional HMT starter, featuring different sound, higher start torque, and an intentionally longer crank time to help build oil/crankcase pressure prior to ignition

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Reality:

This spawned from wanting just make the valve cover red and get rid of the stupid clutch fan and the options spiraled, okay?

Chassis & Other Supporting Products:

HMT Coilovers

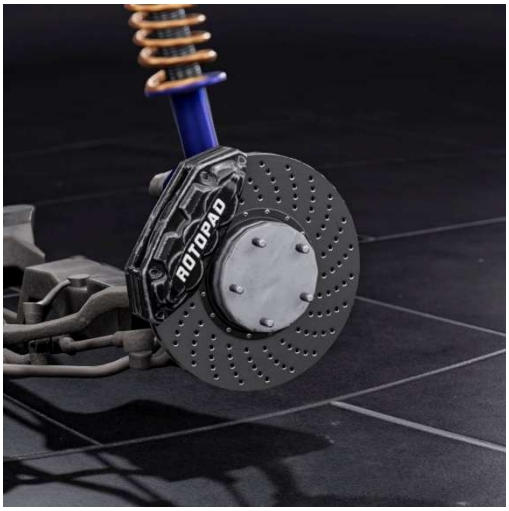


HMT wanted to truly optimize the chassis dynamics of the I-Series, identifying room for improvement over the vanilla coilover options. Hidden within the I-Series files, we discovered an unused, but beautifully detailed, vanilla coilover model. Until now, it hasn't seen any use by the developers or otherwise. By recoloring the coilover body and spring, we gave it a fresh identity to keep it unique when the developers realize they forgot to use the nicer models.

The HMT coilover is a blend of both the Race and Rally coilovers, keeping the lower ride height along with the tighter precompression and stiffer bump stops of the Race coilovers, but adding the dual rate damping from the rally coilovers. This digressive valving is very similar to a lot of much higher end coilovers, like Silvers or OHLINS, allowing for stiffer normal chassis motions but compliance over sudden bumps, keeping the car composed and planted in corners but absorbing sudden harshness like potholes and track curbing.

HMT Helical Limited Slip Differentials

A set of front and rear torque biasing limited slip diff. Inspired by Quaife's ATB differentials along with the diffs in the Sunburst, this design uses machined helical gears rather than clutches and is able to bias torque to the wheel with the most grip, allowing for confident and consistent power into, though, and out of corners. There is no preload harshness or sudden locking and unlocking, just smooth operation on and off power with the thrust forces of the helical gears evenly distributing the power where it's needed most. Offered for both rear and front diffs.



HMT "Track" Brakes

The I-Series receives the full HMT performance upgrade, matching the advanced hardware found on our 800-Series and K-Series builds. This kit represents a significant leap beyond the vanilla race brakes, delivering increased braking force, reduced rotational mass, improved cooling, and a refined, quieter operation.

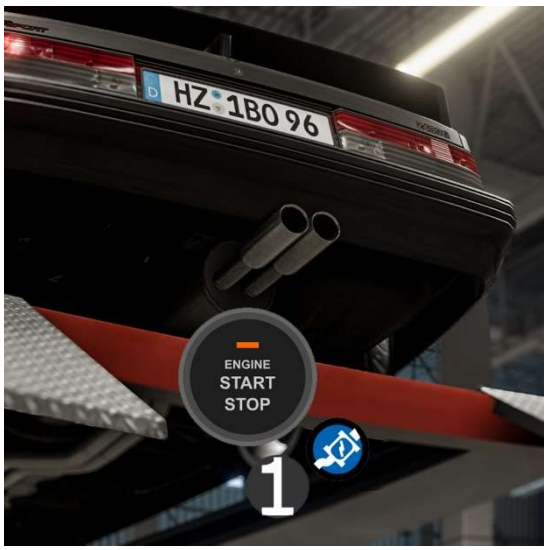
Front and Rear utilize, lightweight two-piece, drilled 320mm rotor and 290mm rotors respectively. These maximize heat dissipation and fade resistance to deliver consistent stopping power. The kit's modern design not only enhances performance but also elevates the visual appeal behind any wheel.

HMT Radiator

Designed firstly as a means of removing the fan shroud that is paired with the OEM clutch fan, which we delete as an option in our engine reconditioning program, It is the next step up from the High-Performance Radiator. Higher volume and efficiency, and when matched with the clutch fan delete kit in our Engine Reconditioning service, is compatible with electronic fans.

HMT Exhaust System

You know what to expect from HMT here now, larger diameter piping, prettier welds, and well though out sound design. Our valved exhaust for the I-Series has a notably different tone than any other option and features a pretty set of chrome shotgun tips extending out the back just a bit more.



For **Patrons**, you get the bonus of a valved variant, by installing the valve controller on the exhaust. The valve allows for a large reduction in sound when desired, the option to choose the default valve position, and features a unique UI Button that doesn't interfere with the map switching. You can change both on the fly, with a trigger enabled button (similar to the map switching) for the interior or using the default key bind, "Alt + E."

Aesthetics

HeckBlende

- Common OEM option of the period, E30-style heckblende, a single-piece rear valance that creates a continuous “mono” taillight and sits behind the license-plate recess.
- Can be found under the “Backlight” slot

Yellow Front Fogs

- Yellow variants of the fog lights
    - Still need to do variants for the Brigsby kit

Taillights with Rear Fog

- Wires in the rear fog light, for which the bulb already exists in the model/glow map

6000 K Headlights

- OEM style Headlights with a brighter, whiter, and more modern halogen drop in bulb rather than the OEM Sealed beam

EUDM Lighting Box

- Under Light color, you’ll see options for Alternate and Standard lights with a “(EUDM)” Suffix, these remove the use of the turn signals as running lights which are mandatory in the US

Paint Matched Airdam

- Paint matched variant of the new for v0.37 base model Airdam, which I find to look better than the ttSport chin lip

HMT Skid Plate

- Similar protection to vanilla skid plate, but with a different model and a lower profile tucked in closer to the sub frame

Modern ETK Changes

Configs:

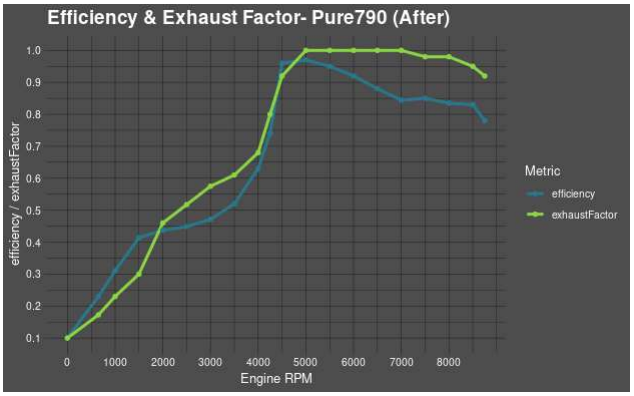
- Adjusted population values in info files, press cars will now be rarer in career mode
  - Required insurance class now Prestige
- Remastered nearly all configs for 2026, see info in vehicle selector for specifics
  - See "Configs" Section for breakdown.

Modern ETKs:

- **NEW: HMT PURE STAGE 1 TORQUE CONVERTER**
  - Made with the "b" suffix kits in mind (Pure430b & Pure560b)
    - Higher stall and quicker shifts out of the 8AT
    - Thank you Nick (BarNonePerformance) for the assist
- **NEW: LAUNCH CONTROL CONFIGURABILITY (PATREON EA)**
  - Part of a minor drive mode rework
    - (and there is much more to come there)
  - Launch control default RPM now user configurable if desired
  - Unique Unicode supplemented interface in tuning menu
- Rework: Pure790 pressure and efficiency curve
  - Drastically smoothed out
  - Output unchanged
  - Spools up and down more predictably
  - Low end torque improved
  - BBG Reworked mildly to match improved delivery
  - *Same power but meaningfully quicker, Cassiopeia Config can run an 8.8 quarter mile at full weight*







- *Fixed HMT TCU Tune Behavior*
  - *Issue Caused shift afterfire to occur on every shift every time regardless of aggression*
  - *Behavior in "D" much more mellow, shift speed slowed for smoothness and refinement*
    - *Behavior in "S" unchanged, shift speed still near instant*

General Changes:

- Took many measures in reducing file size
    - Don't know the exact numbers but simplicity of filter colors has been made much easier and quicker with baseColorFactor and Components/materialOverride
- Built several development tools in R Studio to data mine jbeam blocks during ongoing development of future contents
    - Only reason I caught and was able to fix the Pure790
    - This effects none of you directly, but will help with further optimization
    - If you're a dev that knows how to code in R, you're welcome to reach out for the files
- Fixed sound on etkc Straight pipe
- Removed particulate effect from Eden I & II
- Improved Eden ***Soar*** intake sounds a tad (**Patreon** Only)
- Fixed Missing filters for Eden
- ***Removed Slate Interior color, may get added back***

Patreon Content:

New Patreon Only content:

- Map Switching for ETK I Echo Standalone
  - See Page 11-12
- Exhaust valve for ETK I Echo Exhaust
- Launch Control Configurability (in DSE)





Pre-Existing Patreon Content

HMT Eden II NA Intake:

- Modeled Long Tube Headers
- 570bhp
- 9600RPM Redline
- Optional Soar Edition
  - Mounted behind grille
  - Unique Intake Side Audio File



HMT Carbon Ceramic Brake Pads:

In the brake model in this game, the ABS system often limits threshold braking, which can affect optimal braking distance. These pads provide a solution to this, allowing for additional braking force without having to go through the ABS system.

HMT Twin 2x 8lb Nitrous Bottles:

- 16lbs total with clearance for trunk loads on both the K and 800 platform.
- Unique **silver** Nitrous Bottles

HMT Carbon Fiber Half-shafts:

- Aesthetically pleasing detail and you will have a hard time breaking these.

Inline packages intake cover:

- Add a cover to the heat shielding to hide the filter from view and reduce noise and heat marginally.
- Currently only for 800 series chassis.
  - K series and Pure 860 support in progress

Coal Tune for Dusk packages

- **See page 5 of v2.0.1 Release Notes** for details and explanation:

Colorable Filters:

- Added **colorable filter options,** both **primary**, **secondary** and **tertiary** palettes, to **all** HMT Turbo systems.